VASANTDADA PATIL PRATISHTHAN'S COLLEGE OF

ENGINEERING AND VISUAL ARTS

Department of Artificial Intelligence & Data Science

Academic Year 2026–27

Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

(HiSE — ASL-TR-2026-002)

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Engineering in Artificial Intelligence & Data Science

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2026–27

CERTIFICATE

This is to certify that the project entitled

"Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval (HiSE)"

is submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of Undergraduate in Artificial Intelligence & Data Science.

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide for their invaluable guidance, continuous support, and encouragement throughout the course of this project. We thank the Head of Department and all faculty members of the Department of Artificial Intelligence & Data Science at VPPCOE for providing the necessary infrastructure and academic environment.

We acknowledge Applied Sentience Labs Pvt. Ltd. for providing access to research infrastructure including the ASL Glory server (AMD Ryzen 9 7900X, NVIDIA RTX 5070 Ti 16GB) which serves as the primary training platform for this research. We also thank the open-source community — HuggingFace, PyTorch, and the authors of all cited papers — whose work forms the foundation of this research.

Finally, we thank the University of Mumbai and Vasantdada Patil Pratishthan's College of Engineering & Visual Arts for the opportunity to undertake this research as part of our final year major project.

ABSTRACT

Dense retrieval systems depend on embedding models to produce vector representations that preserve semantic meaning across queries and documents. Current embedding models encode token position using linear mechanisms — predominantly Rotary Position Embedding (RoPE) — which capture sequential distance between tokens but are architecturally blind to hierarchical structural position: the heading path, table row-column coordinate, or JSON key-path that governs a token's contextual meaning in structured documents. This produces a systematic failure mode in which tokens with near-identical surface text but structurally-opposed meanings are assigned dangerously similar embedding vectors, causing retrieval systems to surface structurally incorrect context for downstream language model reasoning.

We present HiSE (Hierarchical Structure-aware Embeddings), a contrastive retrieval embedding model incorporating HiSA (Hierarchical Structural Addressing): a novel positional encoding mechanism that extends RoPE from one-dimensional sequence position to multi-dimensional structural address — encoding depth, parent path, and sibling index as explicit rotational components. HiSE is trained from a small SLM backbone via contrastive learning with structurally-adversarial hard negatives, validated on StructEval-Bench v1 — the first retrieval benchmark explicitly designed to test structural discrimination — and on standard BEIR and MTEB retrieval subsets to confirm structural gains do not degrade general retrieval performance.

Keywords: Rotary Position Embedding, Dense Retrieval, Hierarchical Structural Encoding, RAG Systems, Contrastive Learning, Embedding Models, Structural Discrimination, HiSA, StructEval-Bench

Chapter 1

INTRODUCTION

1.1 Background and Motivation

The global Retrieval-Augmented Generation (RAG) market was valued at $2.33 billion in 2025 and is projected to reach $81.51 billion by 2035, with a CAGR of 42.7% [1]. Document retrieval is the dominant function, accounting for 32.4% of all RAG revenue. Despite this rapid adoption, naive RAG systems achieve only 10–40% success rates in enterprise environments — with retrieval failure, not generation failure, identified as the primary bottleneck [2].

Over 80% of enterprise data is unstructured and hierarchically organized — financial reports, legal contracts, technical documentation, configuration files, and API specifications. Every query against these documents passes through an embedding model. The quality of that embedding model directly determines whether the correct context reaches the downstream language model.

1.2 The Core Problem

Every major embedding model in production — BGE-M3, Qwen3-Embedding, E5-Mistral, Nomic Embed — encodes token position as a single integer via Rotary Position Embedding (RoPE). RoPE computes attention scores as a function of relative sequence distance only: Attention(A,B) ~ f(|pos_A - pos_B|). This architecture is fundamentally blind to structural address — the depth in a document tree, the parent section identity, the sibling index — that governs what a token actually means in hierarchically structured documents.

The consequence is a systematic retrieval failure. The sentence 'Revenue grew 40%' embedded under the heading '## Q3 Performance > Revenue' and the same sentence under '## Q3 Risks > Revenue' produce near-identical embedding vectors in every current production model

(cosine similarity ≈ 0.97). A RAG system querying 'what are the Q3 risks?' may retrieve the performance-framed chunk as equally or more relevant than the risk-framed chunk, causing the downstream language model to generate a confidently wrong answer.

1.3 Research Gap

The benchmarks all production embedding models optimize for — BEIR and MTEB — contain zero structurally-adversarial test pairs. If a benchmark never tests structural sensitivity, no model will optimize for it, and no researcher will detect the failure until it surfaces in a production system. This explains why the gap has persisted despite significant investment in embedding model research through 2023–2025.

1.4 Proposed Solution — HiSE

This project proposes HiSE (Hierarchical Structure-aware Embeddings), incorporating HiSA (Hierarchical Structural Addressing): a novel positional encoding mechanism extending RoPE from 1-dimensional sequence position to 4-dimensional structural address. The core formula changes from $\theta = \theta_1 \times m$ (standard RoPE, one rotation) to $\theta = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$ (HiSA, four rotations), where m = sequence position, d = depth, p = parent section hash, s = sibling index. The project also introduces StructEval-Bench v1 — the first retrieval benchmark designed to measure structural sensitivity.

Chapter 2

REVIEW OF LITERATURE

The following table presents a structured review of prior work most directly relevant to HiSE, covering structure-aware encoding, positional embedding extensions, and retrieval benchmarking:

Paper	Year	What it did	Why it's not enough
TUTA (Microsoft)	2021	Bi-dimensional tree coords for table structure encoding	Classification only — no contrastive retrieval training
DOM-LM	2022	Depth + parent + sibling PE in BERT for HTML docs	HTML-only, additive not rotational, no RAG use case
Tree-Enhanced CodeBERTa	2025	AST structural PE integrated into CodeBERTa	Code/AST only, classification tasks, not a retrieval model
TabBiN (EDBT)	2025	2D tabular coordinates + visibility matrix for tables	Table-specific, no cross-format generalization, no dense retrieval
Struc-EMB (Georgia Tech)	2025	Structural context in LLM-based embedders	Prompt-level workaround — no architectural PE change
RethinkRoPE	2025	N-dimensional RoPE extension blueprint (Lie groups)	Vision/spatial domains only — not applied to documents or retrieval

2.1 Key Observation from Literature

The critical insight from this survey is that the three communities most relevant to HiSE — table understanding (TUTA, TabBiN), document structure encoding (DOM-LM), and dense

retrieval (BGE, Nomic, E5) — have never converged into a single unified system. Table researchers built structure-aware architectures for classification tasks. Retrieval researchers optimized for BEIR/MTEB leaderboard scores. Nobody applied tree-positional encoding to a contrastive retrieval model trained with structurally-adversarial hard negatives and validated on a structural sensitivity benchmark. That is precisely what HiSE does.

Chapter 3

RESEARCH METHODOLOGY

3.1 HiSA — Architectural Contribution

HiSA replaces the single-integer position input to RoPE with a 4-dimensional structural address tuple: (sequence_position, depth, parent_id, sibling_index). Each dimension receives its own learned rotation frequency:

$$\theta(\text{HiSA}) = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$$

Where: m = sequence position, d = depth in document tree (H1=1, H2=2, H3=3, JSON nesting level), p = parent section hash (fixed-size integer ID), s = sibling index among nodes at same level under same parent. Only 96 new scalar parameters are added — surgically small change, structurally large effect.

Critical degenerate case guarantee: when a document has no structure, d=0, p=0, s=0 and HiSA reduces exactly to standard RoPE. This ensures HiSE-Embed loses no performance on plain-text retrieval benchmarks like BEIR and MTEB.

3.2 Phase Plan

- Phase 1 (Months 1–2): Failure-mode survey — quantitatively prove the structural discrimination gap using BGE-M3, Qwen3-Embedding, Nomic Embed on structurally-adversarial document pairs. Establish baseline cosine similarity scores and Retrieval Inversion Rate (RIR).
- Phase 2 (Month 2–3): HiSA implementation — modify RoPE in a small transformer backbone to accept 4D structural addresses. Implement document parsers for markdown, JSON, and tabular formats.

- Phase 3 (Months 3–4): StructEval-Bench v1 construction — build 300–500 structurally-adversarial query-document pairs with ground-truth relevance labels. Define SDS (Structural Discrimination Score) and RIR metrics.
- Phase 4 (Months 4–5): Training and ablation — train HiSE-Embed with structurally-adversarial hard negatives. Run controlled ablation: same backbone, same data, same compute — only positional encoding changes.
- Phase 5 (Month 6): Evaluation and open-source release — validate on StructEval-Bench v1 and BEIR/MTEB. Release all artifacts under asl-research/hise.

3.3 Hardware and Software

- Primary Training: ASL Server 'Glory' — AMD Ryzen 9 7900X, NVIDIA RTX 5070 Ti 16GB VRAM, 32GB DDR5 RAM, Ubuntu 24.04 LTS.
- Development: ROG Strix 'Void' — Intel i7 13th Gen, NVIDIA RTX 4070 8GB VRAM, 16GB DDR5 RAM.
- Framework: Python 3.11+, PyTorch 2.x, CUDA 12.8, HuggingFace Transformers, Sentence-Transformers.
- Evaluation: BEIR evaluation harness, MTEB benchmark, custom StructEval-Bench v1 pipeline.
- Infrastructure: Docker, FAISS vector indexing, Weights & Biases experiment tracking, HuggingFace Hub.

Chapter 4

RESULT ANALYSIS

The following table presents current production model baselines on BEIR nDCG@10 and the projected HiSE-Embed targets. Structural Discrimination and Retrieval Inversion Rate columns are marked 'Not tested' — confirming the benchmark gap that StructEval-Bench v1 fills:

Model	BEIR nDCG@10	Structural Discrimination	Retrieval Inversion Rate
BGE-M3 (570M)	0.467	Not tested	Not tested
Nomic Embed v2 (137M)	0.521	Not tested	Not tested
Qwen3-Embedding-0.6B	0.541	Not tested	Not tested
NV-Embed-v2 (7.8B)	0.627	Not tested	Not tested
HiSE-Embed (target, ≤300M)	**≥ 0.46 ± margin**	**SDS ↑ significant**	**RIR ↓ significant**

4.1 Expected Primary Outcome

HiSE-Embed achieves statistically significant improvement in Structural Discrimination Score (SDS) on StructEval-Bench v1 compared to a same-size, same-data, same-compute standard RoPE baseline. The controlled ablation design — varying only the positional encoding mechanism — ensures this improvement is attributable to HiSA and not data distribution or model scale.

4.2 Expected Secondary Outcome

Retrieval Inversion Rate drops measurably — structurally-adversarial pairs that previously retrieved the wrong chunk with high confidence no longer do so. nDCG@10 on BEIR/MTEB subsets remains within margin of error (±2%), confirming structural gains do not come at the cost of general retrieval performance.

4.3 Benchmark Contribution

StructEval-Bench v1 constitutes a standalone research contribution independent of HiSE-Embed's final numbers. It is the first retrieval benchmark explicitly designed to measure structural sensitivity, and its release enables the broader research community to evaluate any embedding model against this dimension for the first time.

Chapter 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project addresses a fundamental and previously unmeasured architectural limitation in dense retrieval embedding models: the inability to encode hierarchical structural position. The contribution is not a domain-specific fix or a prompt-engineering workaround — it is a targeted modification to the positional encoding mechanism itself, validated through controlled ablation and a purpose-built benchmark.

HiSE does not improve an existing model by training on more data. It corrects the inductive bias: replacing the assumption that position in a sequence is sufficient to represent position in a document. The three research artifacts — HiSA (mechanism), HiSE-Embed (model), StructEval-Bench v1 (benchmark) — each constitute independent, verifiable contributions to the field of information retrieval and representation learning.

5.2 Future Scope

- HiSA → Generative LLMs: Apply the same mechanism to generative model context windows. A structurally-aware LLM would attend differently to tokens in structurally-opposed sections during generation — not just retrieval — potentially improving RAG answer quality beyond what better retrieval alone provides.
- Cross-lingual Structural Retrieval: Extend HiSE to Indic languages where document structure is often the primary disambiguation signal. This is directly relevant to enterprise document search in the Indian market.
- StructEval-Bench v2: Expand the benchmark to legal contracts, clinical trial protocols, and financial 10-K filings with real-world enterprise annotation from domain experts.

- StructRAG Production Integration: Integrate HiSE-Embed into the full ASL DataSense RAG pipeline as StructRAG, enabling production validation against real enterprise document queries.

REFERENCES

No.	Title	Authors	Venue / Year
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[3]	TUTA: Tree-based Transformers for Generally Structured Table Pre-training	Wang, Z. et al.	KDD 2021
[4]	DOM-LM: Learning Generalizable Representations for HTML Documents	Deng, X. et al.	arXiv:2201.10608 2022
[5]	Struc-EMB: The Potential of Structure-Aware Encoding in Language Embeddings	Liu, S. et al.	arXiv:2510.08774 2025
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PUBLICATION DETAILS

Target Conference Publication

Primary: ICNLP 2027 (IEEE Xplore) — International Conference on Natural Language Processing

Submission Deadline: November 30, 2026

Paper Title: HiSE: Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

Backup Venues

- PUNECON 2026 (IEEE Pune Section) — Deadline: September–October 2026
- MLNLP 2026 (IEEE) — ML + NLP focus, confirms topic fit
- ACL/EMNLP 2027 Workshop Track — if results are strong enough post-primary submission

Open Source Release

- HiSE-Embed model — HuggingFace Hub: asl-research/hise-embed (Apache 2.0)
- StructEval-Bench v1 dataset — HuggingFace Datasets + GitHub (CC BY 4.0)
- HiSA implementation and training code — GitHub: asl-research/hise (MIT License)

Note: An arXiv preprint will be submitted simultaneously with the conference submission for visibility. This does not conflict with IEEE submission policies.